



PHARMACEUTICAL DOSAGE FORMS II- PPC 202

SPRING 2025-2026

Lecture (7)

Recent advances in semisolid preparations & Evaluation of semisolid dosage forms



Prepared by

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Intended Learning Outcomes “ILO’s”

After finishing “Lecture 6” the student should be able to:

- Identify **miscellaneous types** of dermal applications
- Define **ideal SSDF**
- Differentiate between **packaging containers** for SSDFs
- Summarize **quality control testing** of semisolid dosage forms

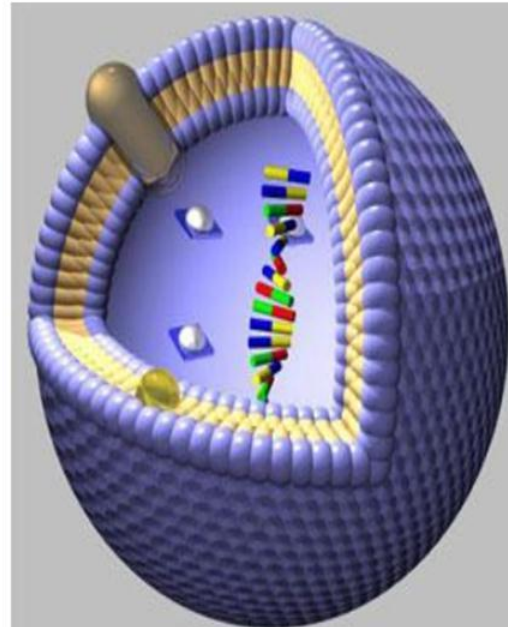
Lecture content:

After finishing “Lecture 6” the student should be able to:

- **Miscellaneous** types of dermal applications
- **Packaging** of semisolid DF
- Recent **advances** in semisolid **preparations**
- Evaluation of semisolid dosage forms (**Quality control testing**)

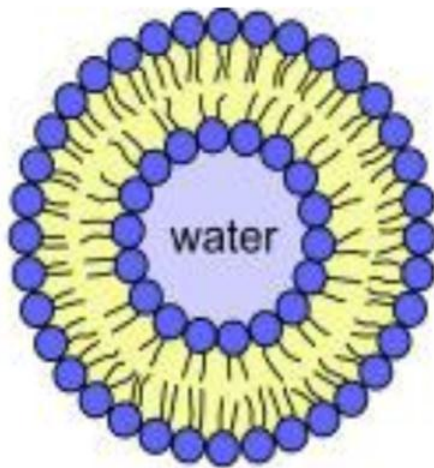
Some Recent advances in semisolids preparations

- Novel semisolids are made up from **water washable bases** so they cause less irritation to skin and are **superior** to **conventional** semisolid dosage form
- Nano-carriers, **e.g. Liposomes** delivers drug in to the dermal & transdermal systems & retained for providing the **prolonged release of drugs**.
- **Liposomes** can serve a **carrier** for both **hydrophilic** and **lipophilic** drugs. Liposomes are claimed to pass through the skin

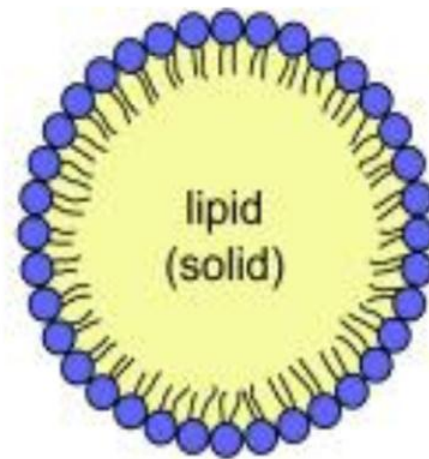


Some Recent advances in semisolids preparations

➤ Solid lipid nanoparticles provide **sustained drug release** , **better penetration** of drugs because: the small particle size of their drug-carrier system ensures close contact to the stratum corneum and increases the amount of encapsulated drug penetrating the skin.



Liposome
(Lipid bilayer enclosing an aqueous core)



Solid lipid nanoparticle (SLN)
(Lipid monolayer enclosing a solid lipid core)

Examples of some advanced topical products on the market:

□ Silver nanoparticles Cream

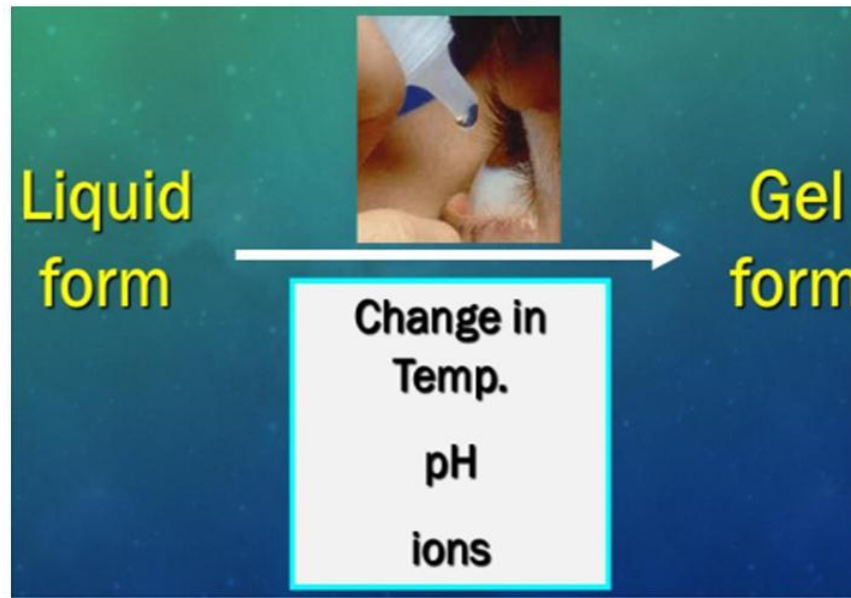
- It has **anti-fungal**, **antibacterial** and **anti-inflammatory** properties. It is used on **cuts**, **burns**, **scratches**, **insect bites**, **fungal rashes**, **psoriasis**, **eczema**, etc.



Phase transition system (in situ gelling system)

□ Formulation of gels depends on factors like

- **Temperature** modulation such as **poloxamers**
- **pH change** such as cellulose acetate phthalate (**CAP**)
- **Presence of ions** such as **Gellan gum**



Packaging of Semisolid Preparations

Describe the packaging material used for SSDF:

An ideal container should

- Protect the product from external atmosphere such as **heat** and **humidity**
- Be non-reactive with product component
- Be easy to use, light in weight and economic

1. Jars

- **Plastic** or opaque **glass** jars some colored green, amber or blue with a cover used with:
- Cosmetic preparation
- **Very thick** products e.g. **pastes** (difficult to extrudate from the tube)



❑ Disadvantages:

1. Bacterial contamination
2. Product oxidation
3. Increase water loss



2. Tubes



- **Plastic** tube such as **High-density polyethylene (HDPE)** & **Polyethylene terephthalate (PET)**
- **Vinyl Lacquered aluminum tube** (epoxy resin lined) to **prevent any interaction** with the product
- **Advantages:**
 1. Decrease water loss , contamination and oxidation
 2. Ease of application



3. Eye ointment container

- Aluminium or plastic tube
- Tube of a capacity of **maximum 5 gm**
- Tube fitted with a **nozzle for application** without **contamination**



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Evaluation of Semisolid Dosage Forms

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Evaluation of Semisolid Dosage Forms

- 1- Visual examination (organoleptic properties)
- 2- Average filling weight
- 3- Homogeneity
- 4- Spreadability
- 5- Determination of pH
- 6- Microbial content
- 7- Rheological studies
- 8- In-vitro drug release
- 9- Content uniformity
- 10- Accelerated stability studies

Evaluation of Semisolid Dosage Forms

1- Visual examination

- The **color** and other **physical** characteristics of the product are noted by visual examination.



Evaluation of Semisolid Dosage Forms

2- Average filling weight

- In this test, the net **weight** or **volume** of the **contents** of the filled containers is determined.
- It is carried out to **ensure** that the product contains the **proper content** when compared with the **labeled claim**.
- Find the **difference** between **filled** and **emptied** container



Evaluation of Semisolid Dosage Forms

3- Homogeneity

- Test for homogeneity of semisolid preparations is done to assure the **absence of gritty particles** in the preparation upon its application to the skin.



Evaluation of Semisolid Dosage Forms

4- Spreadability

- **Spreadability** is the degree of spreading of the semisolid preparation upon its application to the skin.
- A sample of **1 g** of semisolid is placed at the **center** of the **glass plate** (10 x 10 cm) and another glass plate is placed over it carefully above the glass plates 2 kg weight was placed at the center of the plate avoid sliding of the plate. The **diameter** of the semisolid in **cm** is measured **after 30 min**.



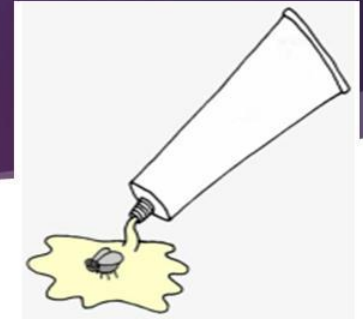
Evaluation of Semisolid Dosage Forms

5- pH determination

- pH of the semisolid preparation should be measured to avoid irritation upon its application to the skin, **using a pH meter**.

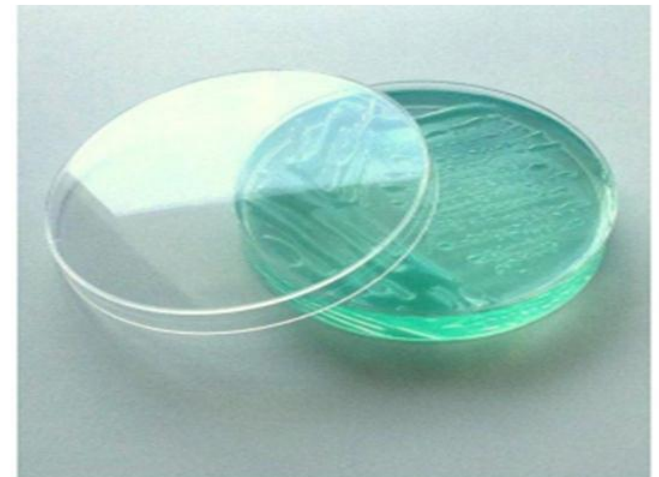


Evaluation of Semisolid Dosage Forms



6- Microbial content

- Topical preparations don't require sterility, except **ocular**.
- They must meet acceptable standards for microbial contents.
- Semisolid preparations containing **water** tend to support microbial growth to a greater extent > than **water free** preparations.



Evaluation of Semisolid Dosage Forms

7- Rheological test:

- The **viscosity** of the semisolid preparation is measured using a Brookfield viscometer



Evaluation of Semisolid Dosage Forms

The following tests are performed for product development

8- Content uniformity

9- In-vitro drug release

10- Accelerated Stability Test

*Thank
you*



END OF TOPICAL SEMISOLID DOSAGE FORMS